

inputs, such as in Computer Vision (CV) applications, representations are often learned through instance classification, in which each image and its augmentations are treated as a single output class to be pulled together [14], [15] or contrasted against other negative samples [22].

Speech signals differ from text and images in that they are *continuous-valued sequences*. Self-supervised learning for the speech recognition domain faces unique challenges from those in CV and NLP. Firstly, the presence of multiple sounds in each input utterance breaks the instance classification assumption used in many CV pre-training approaches. Secondly, during pre-training, there is no prior lexicon of discrete sound units available, as in NLP applications in which words or word pieces are used, hindering the use of predictive losses. Lastly, the boundaries between sound units are not known, which complicates masked prediction pre-training.

In this paper, we introduce **Hidden unit BERT (HuBERT)** that benefits from an offline clustering step to generate noisy labels for a BERT-like pre-training. Concretely, a BERT model consumes masked continuous speech features to predict pre-determined cluster assignments. The predictive loss is only applied over the masked regions, forcing the model to learn good high-level representations of unmasked inputs to infer the targets of masked ones correctly. Intuitively, the HuBERT model is forced to learn both acoustic and language models from continuous inputs. First, the model needs to model unmasked inputs into meaningful continuous latent representations, which maps to the classical acoustic modeling problem. Second, to reduce the prediction error, the model needs to capture the long-range temporal relations between learned representations. One crucial insight motivating this work is the importance of consistency of the targets, not just their correctness, which enables the model to focus on modeling the sequential structure of input data. Our approach draws inspiration from the DeepCluster method for self-supervised visual learning [23]; however, HuBERT benefits from the masked prediction loss over speech sequences to represent their sequential structure.

When the HuBERT model is pre-trained on either the standard Librispeech 960h [24] or the Libri-Light 60k hours [25], it either matches or improves upon the state-of-the-art wav2vec 2.0 [6] performance on all fine-tuning subsets of 10mins, 1h, 10h, 100h, and 960h. We present systematic results on three model sizes pre-trained with HuBERT: BASE (90M parameters), LARGE (300M), and X-LARGE (1B). The X-LARGE model shows up to 19% and 13% relative WER improvement from LARGE models on dev-other and test-other evaluation subsets when pre-trained on the Libri-Light 60k hours.

II. METHOD

A. Learning the Hidden Units for HuBERT

An acoustic model trained on text and speech pairs provides pseudo-phonetic labels for each frame via forced alignment in semi-supervised learning. On the contrary, the self-supervised representation learning setup has access to speech-only data. Nevertheless, simple discrete latent variable models such as

k-means and Gaussian mixture models (GMMs) infer hidden units that exhibit non-trivial correlation with the underlying acoustic units [26] (see also Table V). More advanced systems can achieve better acoustic unit discovery performance using better graphical models [27], [28] or parameterizes the distributions with more powerful neural network models [29]–[33].

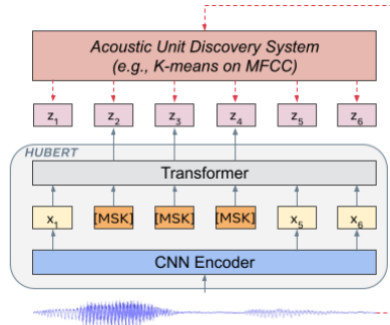


Fig. 1: The HuBERT approach predicts hidden cluster assignments of the masked frames (y_2, y_3, y_4 in the figure) generated by one or more iterations of k-means clustering.

Inspired by this, we propose to use acoustic unit discovery models to provide frame-level targets. Let X denote a speech utterance $X = [x_1, \dots, x_T]$ of T frames. Discovered hidden units are denoted with $h(X) = Z = [z_1, \dots, z_T]$, where $z_t \in [C]$ is a C -class categorical variable and h is a clustering model, e.g. k-means.

B. Representation Learning via Masked Prediction

Let $M \subset [T]$ denote the set of indices to be masked for a length- T sequence X , and $\tilde{X} = r(X, M)$ denote a corrupted version of X where x_t is replaced with a mask embedding $\tilde{x}$ if $t \in M$. A masked prediction model f takes as input $\tilde{X}$ and predicts a distribution over the target indices at each timestep $p_f(\cdot | \tilde{X}, t)$. There are two decisions to be made for masked prediction: *how to mask* and *where to apply the prediction loss*.

Regarding the first decision, we adopt the same strategies used in SpanBERT [34] and wav2vec 2.0 [6] for mask generation, where $p\%$ of the timesteps are randomly selected as start indices, and spans of l steps are masked. To address the second decision, we denote the cross-entropy loss computed over masked and unmasked timesteps as L_m and L_u , respectively. L_m is defined as:

$$L_m(f; X, M, Z) = \sum_{t \in M} \log p_f(z_t | \tilde{X}, t), \quad (1)$$

and L_u is of the same form except that it sums over $t \notin M$. The final loss is computed as a weighted sum of the two

Page Two Only — this text is exclusive to page 2
Used by test_extract_text_first_n_pages.